Software Requirements Specification

for

Veterinary Clinic Management System (VCMS)

Release 1.0

Version 1.0 approved

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Revision History

Name	Date	Reason for Changes	Version
Aryan,Dev	08/10/2026	Initial draft	1.0 draft 1
Aryan,Dev	08/10/2026	Baseline following internal review	1.0 approved

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for Release 1.0 of the Veterinary Clinic Management System (VCMS). The purpose of the system is to digitize and streamline the day-to-day operations of a veterinary clinic, including pet and owner registration, appointment scheduling, medical record keeping, vaccination tracking, prescription management, and billing. This document is intended for use by the project team responsible for designing, implementing, testing, and maintaining the system. Unless otherwise noted, all requirements specified in this document are of high priority and are committed for Release 1.0.

1.2 Document Conventions

Functional requirements are identified with a unique alphanumeric tag consisting of a short feature prefix followed by a sequential number (for example, APT-3 refers to the third requirement under the Appointment Scheduling feature). Requirements written in bold-prefixed form are to be treated as independently verifiable statements. Higher-level requirements stated in Section 4 are assumed to be inherited by the detailed functional requirements listed beneath them unless a more specific priority is stated. The keyword "shall" indicates a mandatory requirement; the keyword "should" indicates a desirable but non-mandatory characteristic.

1.3 Intended Audience and Reading Suggestions

This document is intended for software developers, project managers, quality assurance and testing staff, clinic administrators acting as domain experts, and academic evaluators reviewing the project. Developers and testers should focus on Sections 3 and 4, which contain the detailed interface and functional requirements. Project managers and evaluators may prefer to begin with Sections 1 and 2 for context before proceeding to the system features. Readers interested in data structures should consult Appendix A and Appendix B.

1.4 Product Scope

The Veterinary Clinic Management System is a web-based platform that will allow a veterinary clinic to manage pet registrations, appointments, medical histories, vaccinations, prescriptions, and billing from a single integrated application. The system replaces the clinic's existing paper-based and spreadsheet-based record keeping, reducing scheduling conflicts, improving the accuracy and accessibility of medical histories, and enabling automated reminders for vaccinations and follow-up visits. The system supports the clinic's broader goals of improving patient (animal) care quality, reducing administrative overhead, and enabling data-driven decision making through reporting.

1.5 References

1. Wiegers, Karl. Software Requirements Specification Template, Process Impact, 1999.

2. Wiegers, Karl. Cafeteria Ordering System SRS (Release 1.0), Process Impact, 2002 — used as a reference format for this document.

3. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

4. Veterinary Clinic Management System — Vision and Scope Document (internal project document).

2. Overall Description

2.1 Product Perspective

The VCMS is a new, self-contained web application that will replace the clinic's current manual, spreadsheet, and paper-file-based processes for registering patients, booking appointments, and maintaining medical records. The system is composed of a browser-based front end, an application server that implements business logic, and a relational database that persists clinical and administrative data. The system is expected to evolve across future releases to support multi-branch clinic networks, integration with diagnostic laboratory systems, and online payment gateways.

2.2 Product Functions

At a high level, the VCMS shall allow authorized users to:

- Register and maintain records for pet owners (clients) and their pets.
- Schedule, reschedule, and cancel appointments with veterinarians.
- Create and maintain electronic medical records, including visit notes, diagnoses, and treatment history.
- Record and track vaccination history and generate reminders for upcoming or overdue vaccinations.
- Create and manage prescriptions, including dosage, refills, and pharmacy handoff.
- Generate invoices, record payments, and track outstanding balances.
- Track clinic inventory of medicines and consumable supplies.
- Manage staff accounts, roles, and permissions.
- Produce operational and clinical reports for clinic management.

2.3 User Classes and Characteristics

User Class	Description
Pet Owner (Client)	A member of the public who owns one or more pets registered with the clinic. Clients may book appointments online, view their pet's vaccination and visit history, and view invoices. Clients are assumed to have basic web-browsing skills and no specialized technical training.
Receptionist / Front Desk	A clinic staff member who registers new clients and pets, schedules and manages appointments, checks patients in and out, and processes payments. This is the highest-frequency user class, expected to use the system continuously throughout business hours.
Veterinarian	A licensed veterinary doctor who examines patients, records diagnoses and treatment notes, orders lab work, writes prescriptions, and updates vaccination records. Veterinarians require rapid access to a patient's complete medical history during a consultation.
Veterinary Technician / Nurse	Clinical support staff who assist veterinarians, administer vaccinations and medications, and record vital signs and preliminary observations prior to a consultation.

Clinic Administrator / Manager	Responsible for overseeing clinic operations, managing staff accounts and roles, reviewing financial and operational reports, and configuring clinic-wide settings such as service pricing and appointment slot durations.
System Administrator	A technical staff member (may be an external IT contractor) responsible for installing, configuring, backing up, and maintaining the VCMS software and its underlying infrastructure.

2.4 Operating Environment

OE-1: The VCMS shall operate as a responsive web application accessible from current versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, on both desktop and tablet form factors.

OE-2: The VCMS server components shall operate on a Linux-based server (Ubuntu 22.04 LTS or later) running a supported application runtime and a production-grade web server.

OE-3: The VCMS shall permit access from within the clinic's local network as well as from the public Internet for client-facing features (for example, online appointment booking), subject to authentication.

2.5 Design and Implementation Constraints

CO-1: The system shall be implemented using a three-tier architecture consisting of a web-based front end, a RESTful application programming interface (API) layer, and a relational database.

CO-2: The system shall use a relational database management system (for example, PostgreSQL or MySQL) for the persistence of clinical and administrative records.

CO-3: All data transmitted between the client and server shall be encrypted using TLS 1.2 or higher.

CO-4: The system's design and code shall conform to the project's internal coding and documentation standards.

CO-5: The system shall comply with applicable local data protection and medical record retention regulations.

2.6 User Documentation

UD-1: The system shall provide a searchable, in-application help section describing all major system functions.

UD-2: The system shall provide a guided onboarding walkthrough the first time a new staff account logs in, which the user may skip or revisit later.

2.7 Assumptions and Dependencies

AS-1: The clinic is open for appointments on defined business days and hours, which are configurable by the Clinic Administrator.

AS-2: It is assumed that each pet owner has a unique, verifiable contact method (email address or mobile phone number) for notifications and reminders.

DE-1: The operation of the vaccination and appointment reminder features depends on the availability of a third-party email/SMS gateway service.

DE-2: The operation of the billing feature depends on the clinic's existing chart of accounts and service price list being made available for initial system configuration.

3. External Interface Requirements

3.1 User Interfaces

UI-1: The VCMS shall provide a role-based dashboard that presents each user class with the functions relevant to its role (for example, veterinarians see today's consultations; receptionists see today's appointment calendar).

UI-2: The system shall provide a persistent navigation menu and a contextual help icon on every screen.

UI-3: All screens and workflows shall be operable using the keyboard alone, in addition to mouse and touch input, to support accessibility.

UI-4: The system shall follow a consistent visual style guide across all modules, including color-coded appointment statuses (for example, scheduled, checked-in, in-progress, completed, cancelled).

3.2 Hardware Interfaces

No specialized hardware interfaces have been identified for Release 1.0. Future releases may integrate with barcode scanners for inventory management and card readers for in-clinic payment processing.

3.3 Software Interfaces

SI-1: Notification Gateway

SI-1.1: The VCMS shall communicate with a third-party email/SMS gateway through a programmatic interface to send appointment confirmations, reminders, and vaccination due notices.

SI-1.2: The VCMS shall log the delivery status returned by the notification gateway for each message sent.

SI-2: Payment Processing (future release hook)

SI-2.1: The VCMS shall expose an internal invoicing data model designed to be extendable to an external payment gateway in a subsequent release.

3.4 Communications Interfaces

CI-1: The VCMS shall send an e-mail and/or SMS message to the pet owner confirming a newly booked or modified appointment.

CI-2: The VCMS shall send an e-mail and/or SMS reminder to the pet owner 24 hours prior to a scheduled appointment.

CI-3: The VCMS shall send an e-mail and/or SMS notice to the pet owner when a vaccination becomes due or overdue.

4. System Features

4.1 Pet and Owner Registration

4.1.1 Description and Priority

This feature allows a Receptionist to create and maintain client (pet owner) and patient (pet) records, and allows a Pet Owner to view and update their own contact information. Priority = High.

4.1.2 Stimulus/Response Sequences

Stimulus: Receptionist requests to register a new client and pet. Response: System prompts for client contact details and pet details, then creates the records.

Stimulus: Receptionist requests to add an additional pet to an existing client. Response: System prompts for pet details and links the new pet to the existing client record.

Stimulus: Pet Owner requests to update their contact information. Response: System validates and saves the updated information.

4.1.3 Functional Requirements

REG-1: The system shall allow a Receptionist to create a new client record containing the owner's name, phone number, e-mail address, and physical address.

REG-2: The system shall allow a Receptionist to create a new patient (pet) record containing species, breed, date of birth or estimated age, sex, color/markings, and microchip number (if available), and link it to a client record.

REG-3: The system shall prevent creation of a duplicate client record by checking for an existing record with the same phone number or e-mail address, and shall warn the Receptionist before proceeding.

REG-4: The system shall allow a Receptionist or Clinic Administrator to merge two client records that are identified as duplicates, preserving the full appointment and medical history of both.

REG-5: The system shall allow a Pet Owner to log in to a self-service portal and view or update their own contact information and their pets' basic details.

REG-6: The system shall allow a Receptionist to mark a client or patient record as inactive without permanently deleting it.

REG-7: The system shall maintain an audit trail of all changes made to a client or patient record, including the user who made the change and the timestamp.

4.2 Appointment Scheduling

4.2.1 Description and Priority

This feature allows a Receptionist or Pet Owner to book, reschedule, or cancel appointments with a specific veterinarian for a specific patient, subject to the veterinarian's availability. Priority = High.

4.2.2 Stimulus/Response Sequences

Stimulus: Receptionist or Pet Owner requests to book an appointment. Response: System displays available time slots for the requested veterinarian and service type, and confirms the booking upon selection.

Stimulus: Receptionist requests to reschedule an existing appointment. Response: If the appointment has not yet been checked in, the system allows selection of a new available time slot.

Stimulus: Receptionist or Pet Owner requests to cancel an appointment. Response: If the appointment has not yet been checked in, the system cancels the appointment and frees the time slot.

4.2.3 Functional Requirements

APT-1: The system shall allow a Receptionist to book an appointment by selecting a patient, a veterinarian, a service type, and an available date and time slot.

APT-2: The system shall allow a Pet Owner to request an appointment through the self-service portal, which shall initially carry a status of "Requested" until confirmed by a Receptionist.

APT-3: The system shall display only time slots that fall within the selected veterinarian's configured working hours and that are not already booked.

APT-4: The system shall prevent double-booking of a veterinarian for overlapping time slots.

APT-5: The system shall allow a Receptionist to reschedule an appointment that has a status of "Scheduled" or "Confirmed" to a new available time slot.

APT-6: The system shall allow a Receptionist or Pet Owner to cancel an appointment that has not yet been checked in, and shall record the cancellation reason.

APT-7: The system shall automatically update an appointment's status to "Checked-In" when a Receptionist checks in the patient upon arrival.

APT-8: The system shall display a calendar view of all appointments for a selected veterinarian and date range, color-coded by status.

APT-9: The system shall send an appointment confirmation and a 24-hour reminder per CI-1 and CI-2.

APT-10: If a Receptionist attempts to book an appointment for a patient with an overdue vaccination that is required for boarding or grooming services, the system shall display a warning but shall allow the Receptionist to proceed.

4.3 Electronic Medical Records Management

4.3.1 Description and Priority

This feature allows a Veterinarian or Veterinary Technician to record and retrieve a patient's complete medical history, including visit notes, diagnoses, lab results, and treatment plans. Priority = High.

4.3.2 Stimulus/Response Sequences

Stimulus: Veterinarian opens a checked-in patient's record during a consultation. Response: System displays the patient's full medical history, including past visits, vaccinations, and active prescriptions.

Stimulus: Veterinarian requests to add a new visit record. Response: System prompts for vital signs, examination notes, diagnosis, and treatment plan, and saves the entry to the patient's history upon confirmation.

4.3.3 Functional Requirements

EMR-1: The system shall allow a Veterinarian or Veterinary Technician to create a new medical record entry for a checked-in patient, capturing vital signs (weight, temperature, heart rate), examination notes, diagnosis, and treatment plan.

EMR-2: The system shall display a chronological view of a patient's complete medical history, including all past visits, diagnoses, treatments, vaccinations, and prescriptions.

EMR-3: The system shall allow a Veterinarian to attach lab results or diagnostic images to a medical record entry.

EMR-4: The system shall allow a Veterinarian to flag a medical record entry as requiring a follow-up visit, optionally specifying a suggested follow-up date.

EMR-5: Once a medical record entry has been finalized (signed) by a Veterinarian, the system shall prevent direct editing of that entry; corrections shall be recorded as a linked addendum that references the original entry.

EMR-6: The system shall allow authorized users to search a patient's medical history by date range, diagnosis keyword, or attending veterinarian.

EMR-7: The system shall restrict access to medical records to the Veterinarian, Veterinary Technician, Clinic Administrator roles, and the record's own Pet Owner (read-only), per SE-6.

4.4 Vaccination Tracking and Reminders

4.4.1 Description and Priority

This feature allows clinic staff to record vaccinations administered to a patient and automatically track due dates for future doses or boosters, notifying pet owners in advance. Priority = High.

4.4.2 Stimulus/Response Sequences

Stimulus: Veterinary Technician records a vaccination administered during a visit. Response: System stores the vaccination record and calculates the next due date based on the vaccine's configured schedule.

Stimulus: A vaccination becomes due within the configured reminder window. Response: System sends a reminder notice to the pet owner per CI-3.

4.4.3 Functional Requirements

VAC-1: The system shall allow a Veterinarian or Veterinary Technician to record a vaccination administered to a patient, including vaccine name, lot number, administering staff member, and date administered.

VAC-2: The system shall maintain a configurable vaccine schedule library that defines the recommended interval before the next dose or booster for each vaccine type.

VAC-3: The system shall automatically calculate and display the next due date for each vaccine recorded for a patient.

VAC-4: The system shall send a reminder notification to the pet owner a configurable number of days before a vaccination becomes due (default 14 days).

VAC-5: The system shall flag a patient's record as having an overdue vaccination if the due date has passed without a corresponding new record.

VAC-6: The system shall allow a Receptionist or Clinic Administrator to generate a printable vaccination certificate for a patient.

VAC-7: The system shall allow authorized staff to generate a list of all patients with vaccinations due or overdue within a specified date range.

4.5 Prescription Management

4.5.1 Description and Priority

This feature allows a Veterinarian to create and manage prescriptions for a patient, including dosage instructions, refill limits, and status tracking. Priority = Medium.

4.5.2 Stimulus/Response Sequences

Stimulus: Veterinarian requests to create a new prescription during or after a consultation. Response: System prompts for medication, dosage, frequency, duration, and refill count, and adds it to the patient's active prescriptions.

Stimulus: Pet Owner or Receptionist requests a refill of an active prescription. Response: If refills remain and the prescription has not expired, the system approves the refill and decrements the remaining refill count.

4.5.3 Functional Requirements

RX-1: The system shall allow a Veterinarian to create a prescription for a patient specifying medication name, dosage, frequency, duration, quantity, and number of permitted refills.

RX-2: The system shall display all active and historical prescriptions for a patient on the patient's medical record view.

RX-3: The system shall allow a Receptionist to process a refill request for an active prescription, provided the remaining refill count is greater than zero and the prescription has not expired.

RX-4: The system shall prevent processing of a refill for an expired or fully consumed prescription and shall prompt the Receptionist to route the request to a Veterinarian for renewal.

RX-5: The system shall decrement the corresponding medication's inventory quantity, per INV-3, whenever a prescription is dispensed from clinic stock.

RX-6: The system shall allow a Veterinarian to discontinue an active prescription, recording the reason for discontinuation.

4.6 Billing and Invoicing

4.6.1 Description and Priority

This feature allows a Receptionist to generate invoices for services rendered and products dispensed, record payments, and track outstanding balances. Priority = High.

4.6.2 Stimulus/Response Sequences

Stimulus: Receptionist requests to generate an invoice for a completed appointment. Response: System compiles all billable services, medications, and products associated with the visit and calculates the total amount due.

Stimulus: Receptionist records a payment against an invoice. Response: System updates the invoice balance and, if fully paid, marks the invoice as "Paid."

4.6.3 Functional Requirements

BILL-1: The system shall automatically compile a draft invoice from the services performed, medications dispensed, and products sold during a completed appointment.

BILL-2: The system shall allow a Receptionist to add, remove, or adjust line items on a draft invoice before it is finalized.

BILL-3: The system shall calculate applicable taxes and the total amount due based on clinic-configured tax rates.

BILL-4: The system shall allow a Receptionist to record a full or partial payment against an invoice, specifying the payment method (cash, card, or other).

BILL-5: The system shall mark an invoice as "Paid" when the recorded payments equal or exceed the total amount due, and as "Partially Paid" or "Unpaid" otherwise.

BILL-6: The system shall allow a Receptionist or Clinic Administrator to generate a printable or downloadable PDF copy of any invoice.

BILL-7: The system shall allow a Pet Owner to view their invoice history and outstanding balance through the self-service portal.

BILL-8: The system shall allow a Clinic Administrator to issue a refund or credit note against a previously paid invoice, recording the reason.

4.7 Inventory Management

4.7.1 Description and Priority

This feature allows clinic staff to track stock levels of medications and consumable supplies, and to receive alerts when stock runs low. Priority = Medium.

4.7.2 Stimulus/Response Sequences

Stimulus: A prescription is dispensed or a consumable item is used during a visit. Response: System decrements the corresponding inventory item's quantity on hand.

Stimulus: An inventory item's quantity on hand falls below its configured reorder threshold. Response: System flags the item as low stock and notifies the Clinic Administrator.

4.7.3 Functional Requirements

INV-1: The system shall maintain a catalog of medications and consumable supplies, each with a name, unit of measure, unit cost, quantity on hand, and reorder threshold.

INV-2: The system shall allow a Clinic Administrator to record incoming stock (purchases or restocking) against a catalog item, increasing its quantity on hand.

INV-3: The system shall automatically decrement an item's quantity on hand whenever it is dispensed via a prescription or consumed during a recorded procedure.

INV-4: The system shall flag an inventory item as "Low Stock" whenever its quantity on hand falls at or below its configured reorder threshold, and shall display these items on the Clinic Administrator's dashboard.

INV-5: The system shall allow a Clinic Administrator to generate a report of inventory usage over a specified date range.

4.8 Staff and Role Management

4.8.1 Description and Priority

This feature allows a Clinic Administrator to create staff accounts, assign roles, and control access permissions across the system. Priority = Medium.

4.8.2 Stimulus/Response Sequences

Stimulus: Clinic Administrator requests to create a new staff account. Response: System prompts for the staff member's details and role assignment, then creates the account and sends login credentials.

4.8.3 Functional Requirements

STAFF-1: The system shall allow a Clinic Administrator to create a staff account with a name, contact information, and an assigned role (Receptionist, Veterinarian, Veterinary Technician, or Clinic Administrator).

STAFF-2: The system shall restrict each user's access to the functions and data permitted for that user's assigned role, per SE-4.

STAFF-3: The system shall allow a Clinic Administrator to deactivate a staff account, immediately revoking that account's access.

STAFF-4: The system shall allow a Clinic Administrator to configure each veterinarian's standard working hours, used by the Appointment Scheduling feature.

STAFF-5: The system shall require a staff member to change their initial system-generated password upon first login.

4.9 Reporting and Analytics

4.9.1 Description and Priority

This feature allows a Clinic Administrator to generate operational and clinical reports summarizing appointments, revenue, and patient statistics. Priority = Low.

4.9.2 Stimulus/Response Sequences

Stimulus: Clinic Administrator requests a report for a specified date range and report type. Response: System compiles and displays the requested report, with an option to export it.

4.9.3 Functional Requirements

RPT-1: The system shall allow a Clinic Administrator to generate a revenue report summarizing invoiced and collected amounts over a specified date range, broken down by service category.

RPT-2: The system shall allow a Clinic Administrator to generate an appointment volume report showing the number of appointments per veterinarian and per service type over a specified date range.

RPT-3: The system shall allow a Clinic Administrator to generate a patient demographics report summarizing active patients by species and breed.

RPT-4: The system shall allow any generated report to be exported in CSV or PDF format.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

PE-1: The system shall support at least 50 concurrent staff users and 200 concurrent Pet Owner portal sessions without degradation of response time.

PE-2: All web pages generated by the system shall be fully loaded within 3 seconds under normal network conditions (broadband or Wi-Fi).

PE-3: Search queries against patient or client records shall return results within 2 seconds for a database of up to 100,000 patient records.

PE-4: The system shall display a confirmation message to the user within 2 seconds after a form submission.

5.2 Safety Requirements

SA-1: The system shall display a clearly visible warning when a Veterinarian attempts to prescribe a medication that is flagged as contraindicated for the patient's recorded species or known allergies.

SA-2: The system shall prevent the deletion of any finalized medical record entry to preserve the integrity of the clinical history.

5.3 Security Requirements

SE-1: All network transactions involving personal, financial, or medical information shall be encrypted in transit using TLS 1.2 or higher, per CO-3.

SE-2: Users shall be required to authenticate with a unique username and password to access any function other than the public appointment-request form.

SE-3: Passwords shall be stored using a salted, industry-standard one-way hashing algorithm; plaintext passwords shall never be stored or logged.

SE-4: The system shall enforce role-based access control such that each user may perform only the functions permitted for their assigned role, per STAFF-2.

SE-5: The system shall automatically terminate an inactive staff session after 20 minutes of inactivity.

SE-6: The system shall permit a Pet Owner to view only the records of pets linked to their own client account, and shall not expose other clients' data.

SE-7: The system shall maintain an audit log of all access to and modification of medical records, retained for a minimum of one year.

5.4 Software Quality Attributes

QA-1: Availability: The system shall be available to staff users 99.5% of the time during the clinic's configured business hours.

QA-2: Reliability: If the connection between a user and the system is interrupted before a form submission completes, the system shall preserve the in-progress data and allow the user to resume upon reconnection.

QA-3: Usability: A new Receptionist with basic computer literacy shall be able to book an appointment without assistance after completing the onboarding walkthrough described in UD-2.

QA-4: Maintainability: The system's codebase shall be organized into clearly separated modules corresponding to the system features in Section 4, to facilitate independent maintenance and testing.

QA-5: Portability: The system's application layer shall not depend on any operating-system-specific features, to allow future migration between hosting environments.

5.5 Business Rules

BR-1: Only staff members with the Veterinarian role may create, finalize, or discontinue a prescription.

BR-2: Only staff members with the Clinic Administrator role may deactivate a staff account or issue a refund.

BR-3: An appointment may not be booked in a time slot that falls outside the assigned veterinarian's configured working hours.

BR-4: A medical record entry becomes immutable once signed by the attending Veterinarian; subsequent corrections must be recorded as addenda, per EMR-5.

BR-5: An invoice may not be marked "Paid" unless the total recorded payments equal or exceed the invoice total.

6. Other Requirements

The system shall retain medical records for a minimum period as required by applicable local veterinary practice regulations, even after a client or patient record has been marked inactive. The system's user interface text shall be maintained in a way that supports future localization into additional languages without requiring code changes. Database backups shall be performed on a daily automated schedule, with backup files retained for a minimum of 30 days.

Appendix A: Glossary

Term	Definition
Patient	A pet registered in the system that receives veterinary care.
Client	A pet owner registered in the system, associated with one or more patients.
Consultation	A single clinical encounter between a Veterinarian and a patient during a checked-in appointment.
EMR	Electronic Medical Record — the digital record of a patient's clinical history.
Vaccine Schedule	A configurable set of rules defining the recommended interval between doses of a given vaccine.
Refill	An additional dispensing of a previously prescribed medication under the same prescription.
Draft Invoice	An automatically compiled, editable invoice awaiting finalization by a Receptionist.
Reorder Threshold	The quantity-on-hand level at which an inventory item is flagged as low stock.
Self-Service Portal	The client-facing area of the system where Pet Owners may view records and request appointments.
Role-Based Access Control (RBAC)	A security model that restricts system access based on a user's assigned role.

Appendix B: Analysis Models

B.1 Data Dictionary

Data Element	Composition
client	= client ID + client name + phone number + e-mail + address + status
patient	= patient ID + client ID + name + species + breed + date of birth + sex + microchip number + status
appointment	= appointment ID + patient ID + veterinarian ID + service type + date + time slot + status
medical record entry	= entry ID + patient ID + appointment ID + veterinarian ID + vital signs + notes + diagnosis + treatment plan + signed flag + timestamp
vaccination record	= record ID + patient ID + vaccine name + lot number + date administered + administering staff ID + next due date
prescription	= prescription ID + patient ID + veterinarian ID + medication + dosage + frequency + duration + quantity + refills remaining + status
invoice	= invoice ID + client ID + appointment ID + 1:m{line item} + tax amount + total amount + amount paid + status
line item	= description + quantity + unit price + line total
inventory item	= item ID + name + unit of measure + unit cost + quantity on hand + reorder threshold
staff account	= staff ID + name + contact information + role + working hours + active flag

B.2 Conceptual Data Model

A Client may own one or more Patients (1:m). A Patient may have many Appointments over time, each associated with exactly one Veterinarian (staff account) and, upon completion, one Invoice. A Patient may accumulate many Medical Record Entries, Vaccination Records, and Prescriptions, each linked back to the Patient and to the staff member who created it. An Invoice is composed of one or more Line Items, which may reference services rendered, prescriptions dispensed, or Inventory Items consumed. This structure mirrors the entity relationships illustrated in Figure 1.

Figure 1 — Entity Relationship Summary

Client (1) —— owns —— (m) Patient Patient (1) —— has —— (m) Appointment —— attended by —— (1) Staff (Veterinarian) Patient (1) —— has —— (m) Medical Record Entry / Vaccination Record / Prescription Appointment (1) —— generates —— (1) Invoice —— composed of —— (m) Line Item Inventory Item (1) —— consumed by —— (m) Line Item / Prescription

B.3 Appointment Status State-Transition Model

An appointment progresses through the following states: Requested → Scheduled/Confirmed → Checked-In → In-Progress → Completed. An appointment may transition to Cancelled from any state

prior to Checked-In. A Completed appointment automatically triggers generation of a Draft Invoice, per BILL-1.

Figure 2 — Appointment State-Transition Summary

Requested --(Receptionist confirms)--> Scheduled Scheduled --(Patient arrives)--> Checked-In CheckedIn --(Veterinarian begins consult)--> In-Progress In-Progress --(Veterinarian finalizes visit)--> Completed --(auto)--> Draft Invoice Created Requested/Scheduled --(cancellation)--> Cancelled

Appendix C: To Be Determined List

1. TBD-1: Final selection of the third-party e-mail/SMS notification gateway provider (see SI-1).

2. TBD-2: Exact medical record retention period required under applicable local veterinary practice regulations (see Section 6).

3. TBD-3: Whether Release 1.0 will include integration with an external payment gateway for online invoice payment, or whether this is deferred to Release 2.0 (see SI-2).

4. TBD-4: Specific list of vaccine types and default schedules to pre-populate the vaccine schedule library (see VAC-2).